



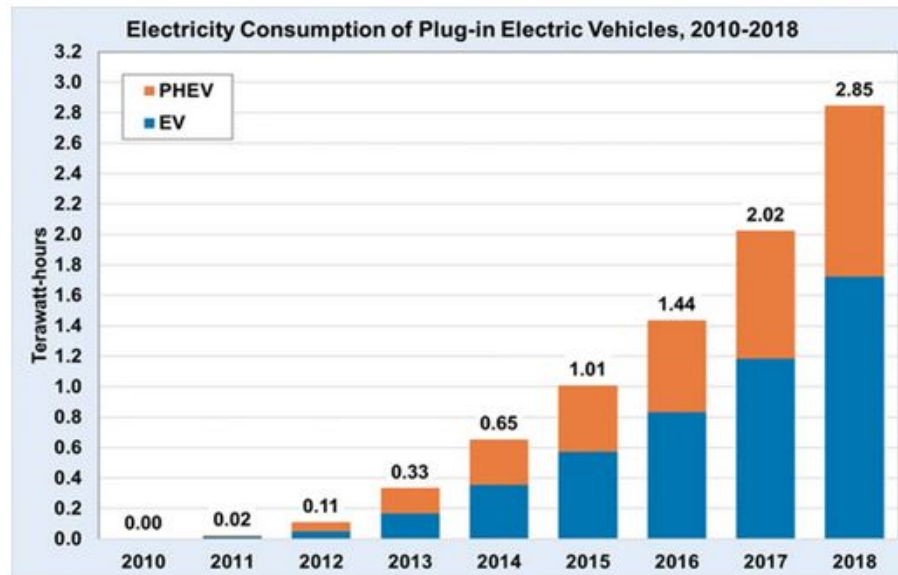
QUANTUM-READY GRID ANALYTICS PLATFORM

Powering Southeast Asia's Energy Transition with Quantum Computing

Seed Round 2026

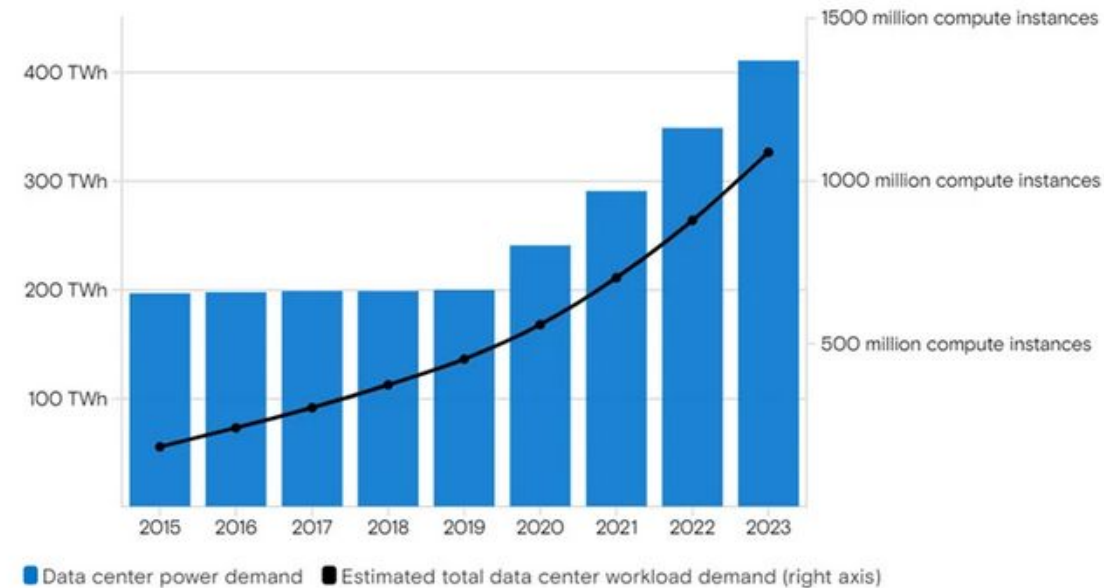
The state of energy consumption

Electric vehicles



DATA CENTER

The workload demand for data centers...
...and the power they consumed

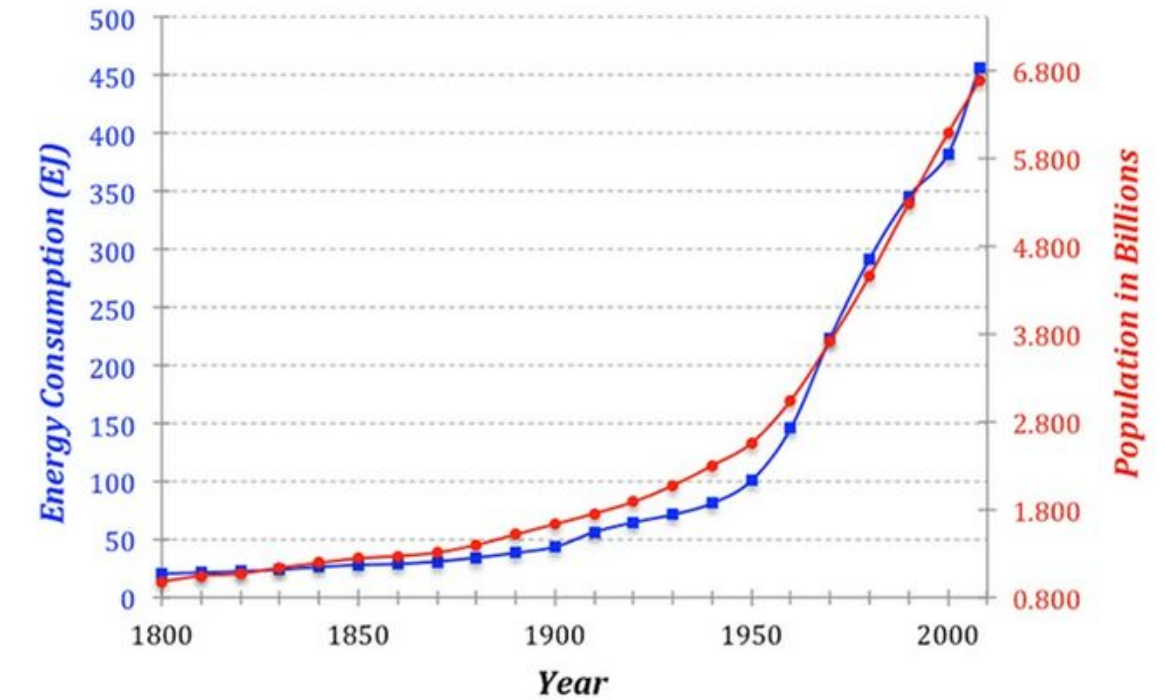


Source: Masanet et al. (2020), Cisco, IEA, Goldman Sachs Research
The data center power demand for 2023 is an estimate.

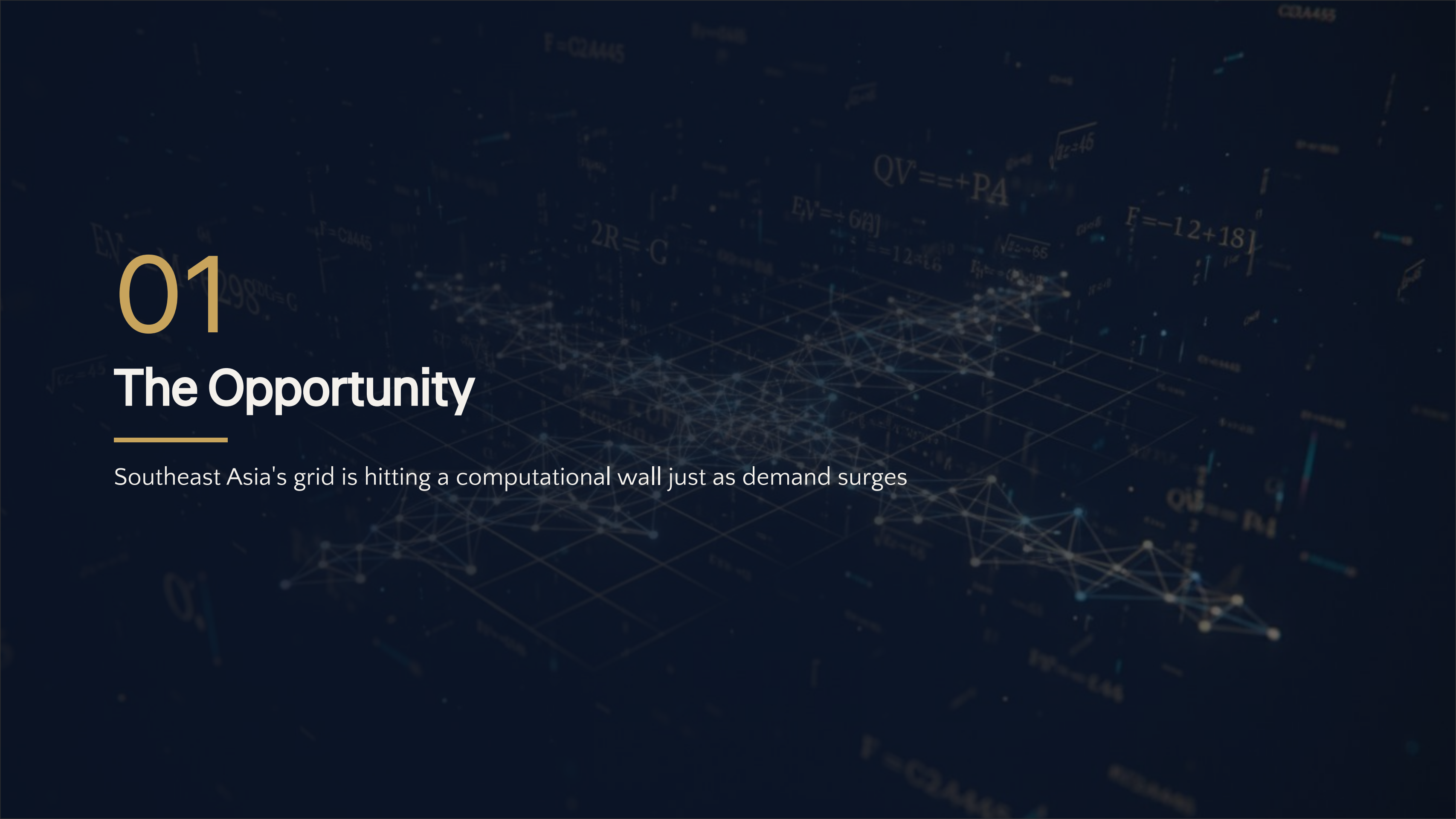
Goldman
Sachs

Population

Energy Consumption and Population



Exponential growth !

The background features a dark blue gradient with a complex network of white and light blue lines and dots, resembling a data or neural network. Overlaid on this are various mathematical formulas in a light blue, slightly blurred font. Visible formulas include $F = C2A45$, $QV = +PA$, $EV = +6A$, $2R = C$, $F = -12 + 18$, $\sqrt{12+45}$, $\sqrt{12+65}$, and $EV = -12 + 18$.

01

The Opportunity

Southeast Asia's grid is hitting a computational wall just as demand surges

Grid Demand Surging 54% by 2035, Straining Classical Compute

2,000+

TWh projected electricity demand by 2035 (IEA)

54%

Demand growth from 1,300 TWh today, driven by cooling + renewables

99,934

MW max capacity target in Vietnam PDP8 by 2030

The Challenge

The IEA projects electricity demand in Southeast Asia rising from **over 1,300 TWh today to above 2,000 TWh by 2035**.

Building cooling and renewables are the two strongest demand drivers, with **renewables supplying more than half of demand growth**.

Vietnam — the region's largest renewable power market — sets aggressive PDP8 targets implying tighter operations and heavier reliance on simulation.

The Bottleneck

Classical grid solvers are hitting computational walls as system complexity grows.

Real-time contingency analysis, state estimation, and optimal power fload require **large sparse linear solves** that become prohibitively expensive at scale.

Grid operators need **10-100x speedups** to manage the renewable transition reliably.



02

Our Solution

A quantum-ready platform purpose-built for power grid analytics at scale

Four Quantum-Enabled Applications, One Linear Solver Kernel



Quantum Power Flow (QPF)

Faster power flow calculations enabling high-volume contingency screening and renewable integration studies



Quantum State Estimation (QSE)

Faster and more robust real-time grid visibility from SCADA/PMU measurements



Quantum Transient Analysis (QEMTP)

Accelerated time-domain simulation for protection systems and inverter-heavy dynamics



Quantum Optimal Power Flow (QOPF)

Accelerating dispatch, congestion management, and market operations



The Unified Kernel Insight

All four workloads repeatedly depend on **large sparse linear solves** — improving that single kernel improves the entire operational toolchain. This unified approach maximizes R&D leverage, reduces time-to-market, and creates a compounding competitive moat as each application strengthens the shared solver.

Technical Roadmap: Current noisy Quantum Computers (NISQ) Today to Noisless Quantum Computer (FTQC) Tomorrow

SHORT-TERM

VQLS (NISQ Era)

Variational Quantum Linear Solver (VQLS) runs on near-term quantum hardware with classical optimization loops. Hybrid quantum-classical approach maximizes utility of today's NISQ devices.

Current focus: QPF prototype with speedup validated on IEEE test systems via Qulacs simulation.

LONG-TERM

HHL (FTQC)

Harrow-Hassidim-Lloyd (HHL) algorithm delivers exponential speedup on mature fault-tolerant quantum computers for solving linear systems.

Target: Full quantum advantage on national-scale grid systems (1,000+ buses) by Year 4-5.

Bridge Strategy

Our simulation stack is built on **Qulacs** (fast Python/C++ quantum simulator), enabling rapid iteration on circuit designs. We combine **Parity Check + preconditioning + ML-based encoding reduction** to create a practical pathway from NISQ to FTQC — already demonstrated outperformance vs. numpy.linalg in end-to-end benchmarks.

03

Technology

Parity Check — our breakthrough that makes quantum linear solvers practical



Parity Check: $O(1)$ Measurement Setup resolves the Quantum Bottleneck

PROBLEM

Measurement Overhead Destroys Speedups

The #1 blocker for quantum linear solvers in power systems is **measurement/readout overhead**.

Traditional approaches require $O(N)$ setup measurements to extract solution coefficients.

This overhead overwhelms the theoretical quantum advantage.



BREAKTHROUGH

Parity Check: $O(1)$ Extraction

Our Parity Check method extracts the **power data state with $O(1)$** measurement setup complexity.

Instead of thousands of measurement setups, we reduce the delay and classical data post-processing time to extract solution in **constant time**.

This is a step-change that makes quantum advantage practical.

Complete Technical Advantage Stack



Parity Check

$O(1)$ measurement extraction of quantum state coefficients, eliminating the measurement bottleneck



Preconditioning

Reduces circuit depth and qubit requirements, making problems tractable on near-term hardware



ML Encoding

Machine learning reduces data input and structures quantum encoding complexity for efficient circuit compilation

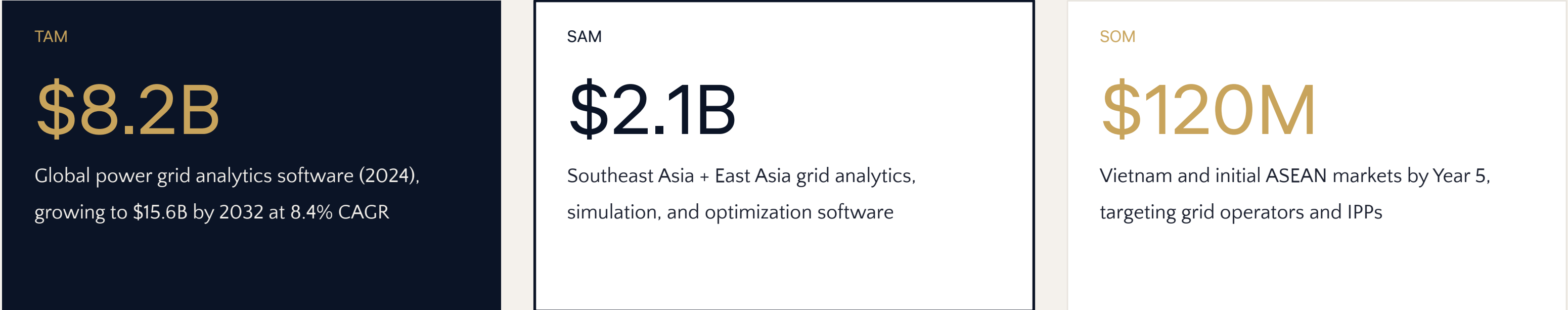
04

Market & Business

A \$4B+ addressable market growing at 15% CAGR, driven by renewable grid integration



TAM/SAM/SOM — Grid Analytics Fueled by Renewable Transition



Key Growth Drivers

- 01 Renewable integration mandates across ASEAN
- 02 Grid congestion from distributed generation
- 03 Regulatory pressure for real-time grid visibility
- 04 Quantum computing maturity converging with grid investment cycles

Source: [Grand View Research, Power Grid Analytics 2024](#)

SaaS Platform + Professional Services + Quantum-as-a-Service

TIER 1 SaaS Subscription \$50K-200K/yr Cloud-based access to QPF, QSE, and QOPF modules. Annual licenses for grid operators and utilities with tiered features.	TIER 2 Professional Services \$300K-1M Custom integration, training, and consulting for national grid operators. High-touch deployment with dedicated support.	TIER 3 · FUTURE Quantum-as-a-Service \$0.10-1.00/call Pay-per-simulation on quantum hardware via cloud API. Scalable usage model for high-volume operational workloads.
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Go-to-Market Strategy

Target customers: National grid operators (EVN Vietnam, EGAT Thailand, PLN Indonesia), independent power producers (IPPs), renewable developers, and system integrators.

Go-to-market: Direct sales to early-adopter utilities → strategic partnerships with Siemens, GE, Hitachi Energy → channel sales through system integrators.

Partnership Stack: **RIKEN / Fujitsu:** Quantum hardware access + credibility | **Qulacs / QunaSys:** Domestic Japan software for quantum simulator ecosystem | **ASEAN Centre for Energy:** Policy alignment for ASEAN Power Grid standards

Development Roadmap: From Lab to Grid Operations



Key Milestones

Month 6 QPF prototype validated

Month 9 5x speedup demonstrated

Month 12 First paying pilot customer

Month 18 Series A ready

Year 2 Commercial SaaS launch

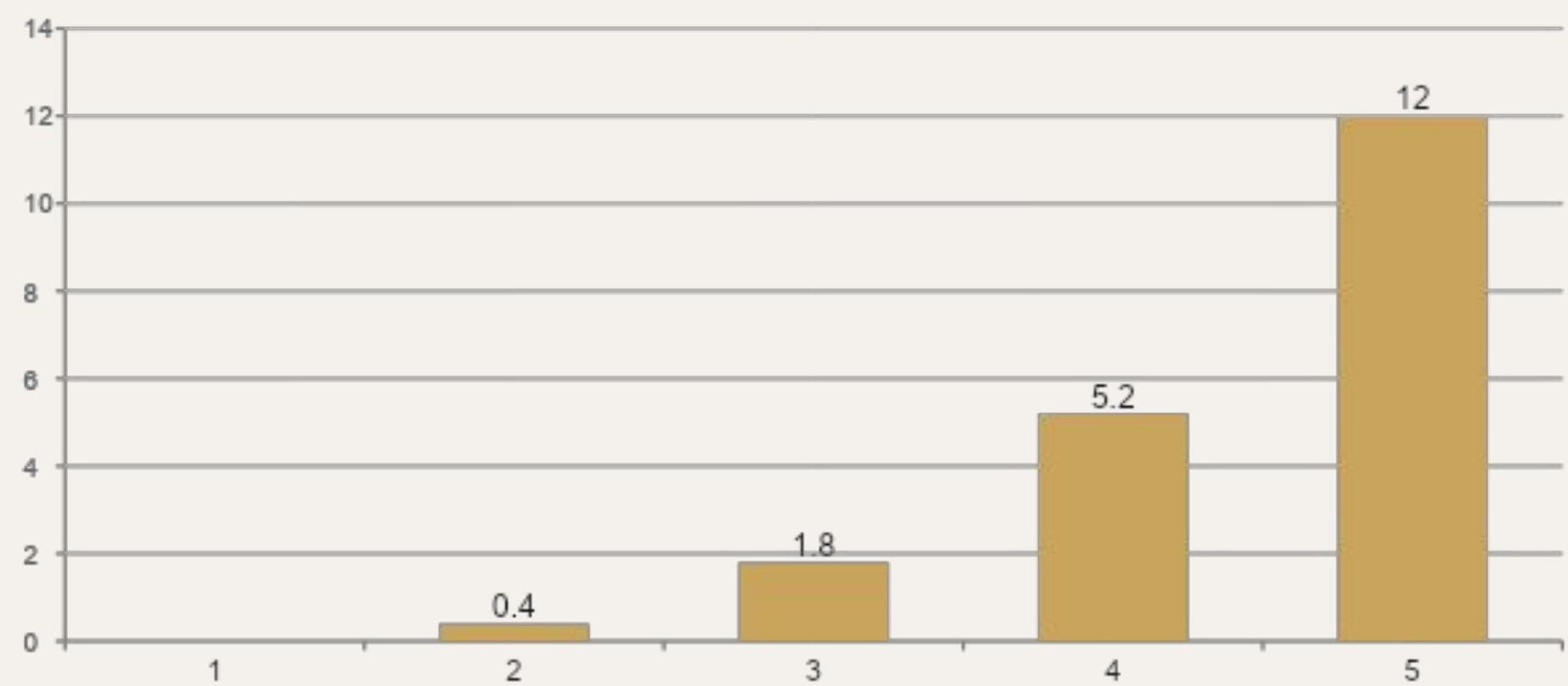
Year 3 20+ customers, \$5.2M

Year 4 QEMTP + HHL transition

Year 5 \$12M revenue, Series B

Year 5+ Full quantum advantage

\$12M Revenue by Year 5, 85% Gross Margin SaaS Business



85%
Gross Margin

19x
LTV / CAC

\$45K
CAC (Enterprise)

\$850K
Customer LTV

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$0	\$400K	\$1.8M	\$5.2M	\$12.0M
Customers	0	2	8	20+	40+
Milestone	R&D	Pilots	Breakeven	Scale	Series B



05

Team & Funding

World-class quantum scientists meets deep grid domain expertise

Leadership: Quantum Research + Engineering + Product



Le Duc Truyen — CEO & Founder

PhD at **RIKEN, Japan**. Quantum computing researcher with deep expertise in variational algorithms and power system applications.



Nguyen Xuan Trieu — CTO

Master's at the **University of Technology, Vietnam National University**. Grid systems engineering and software architecture expert.



Dinh Nhu Duc — Co-CTO

Universite Grenoble Alpes, France. Quantum software and high-performance computing specialist.



Li Jingyang — CPO

PhD at **RIKEN, Japan**. Product strategy, quantum-classical integration, and go-to-market execution.



Scientific Supervisor

Masazumi Honda — Senior Scientist at **RIKEN**. Leading researcher in quantum information and theoretical physics.

Combined strength: 30+ years quantum research experience, direct relationships with Vietnam's power sector, unique positioning at the intersection of quantum computing and energy systems.

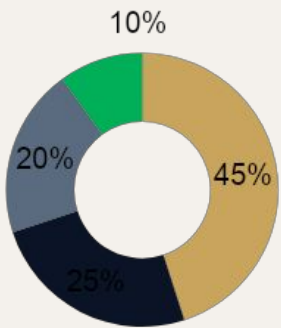
Seed Round: Raising \$1.5M to Reach First Commercial Deployment

SEED ROUND

\$1.5M

18-month runway to Series A readiness

Use of Funds



- R&D
- Pilot
- Team
- Ops

- 45% R&D – Algorithm development, platform engineering
- 25% Pilot – Customer integration, validation
- 20% Team – 4 engineers + 2 sales hires
- 10% Operations – Legal, finance, admin

Key Milestones With This Round

M9

Working QPF/QSE prototype with practical deployment

M12

First paying pilot customer contracted

M12

Patent portfolio filed (2–3 core patents)

M18

Platform ready for Series A (\$8–10M)

Investment Thesis

First-mover in quantum grid analytics | **Massive market** driven by renewable transition | **Defendable IP** with patent-pending Parity Check | **Experienced team** from top research institutions | **Clear path** from NISQ revenue to Fault Tolerance quantum advantage



Thank You

Quantum-Ready Grid Analytics Platform

Let's power the future of energy together

ductruyen.le@riken.jp

SEED ROUND 2026
